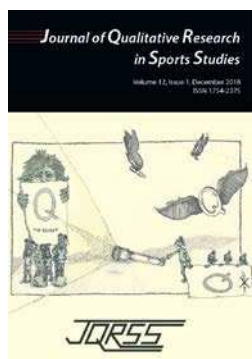


**Published by:**

Sport and Wellbeing Press

University of Central Lancashire, Preston, UK.

**Journal of Qualitative Research in Sports Studies****Volume 12, Issue 1, December 2018****An evaluation of the coaching methods of the batting backlift technique in cricket**<sup>1</sup> Habib Noorbhai (University of Fort Hare, South Africa)<sup>2</sup> Timothy D. Noakes (University of Cape Town, South Africa)**ISSN:** 1754-2375**ISBN:** 978-0-9955744-3-4 (386 pages)**JQRSS Article No:** 2/16-12-1-2018-PG[67]-112**To cite this article:**

Noorbhai, H. and Noakes, T.D. (2018) An evaluation of the coaching methods of the batting backlift technique in cricket.

*Journal of Qualitative Research in Sports Studies*, 12, 1, 35-56.**Self-archived URL link to this article:**[https://www.academia.edu/37841701/Habib\\_Noorbhai\\_and\\_Tim\\_Noakes\\_2018\\_An\\_evaluation\\_of\\_the\\_coaching\\_methods\\_of\\_the\\_batting\\_backlift\\_technique\\_in\\_cricket](https://www.academia.edu/37841701/Habib_Noorbhai_and_Tim_Noakes_2018_An_evaluation_of_the_coaching_methods_of_the_batting_backlift_technique_in_cricket). *Journal of Qualitative Research in Sports Studies* 12 1 35-56**Advice to submitters - see JQRSS Guide to Contents & Open Call for Papers:**[https://www.academia.edu/3513281/JQRSS\\_Overview\\_Guide\\_to\\_Contents\\_and\\_Editorials\\_by\\_Volume\\_-\\_Open\\_Call\\_for\\_Papers](https://www.academia.edu/3513281/JQRSS_Overview_Guide_to_Contents_and_Editorials_by_Volume_-_Open_Call_for_Papers)**JQRSS Mapping of Articles (2007- to date):**[https://www.academia.edu/8827414/Mapping\\_of\\_JQRSS\\_articles\\_across\\_Undergraduate\\_and\\_Postgraduate\\_research](https://www.academia.edu/8827414/Mapping_of_JQRSS_articles_across_Undergraduate_and_Postgraduate_research)

Copyright © Clive Palmer and the individual authors

**Notice:**

The discussions, statements of fact and opinions contained in the articles of The Journal of Qualitative Research in Sports Studies are those of the respective authors and cited contributors and are set out in good faith for the general guidance of student supported research and the promotion of pedagogical discussion in teaching and learning contexts. No liability can be accepted by the Editor, Advisory Board, the reviewers or the authors/submitters for loss or expense incurred as a result of relying upon particular statements made or circumstances outlined in this journal.

**Online – Open Access Research Profiles:**academia.edu: <https://uclan.academia.edu/ClivePalmer>ResearchGate: [http://www.researchgate.net/profile/Clive\\_Palmer](http://www.researchgate.net/profile/Clive_Palmer)British Conference of Undergraduate Research <http://bcu.org/journals/>

# An evaluation of the coaching methods of the batting backlift technique in cricket

**Habib Noorbhai<sup>1</sup> and Timothy D. Noakes<sup>2</sup>**

(University of Fort Hare, South Africa<sup>1</sup>)

(University of Cape Town, South Africa<sup>2</sup>)

*Keywords: Mixed methods, cricket, batting technique, elite performance*

## Abstract

Since the inception of the game, cricket coaches have been coaching batting techniques to cricketers at various levels using mainly cricket coaching manuals. Most of these manuals advocate the straight batting backlift technique (SBBT). However, previous research has shown that the practice of elite batsmen does not match the theory of the backlift. This raises questions about the extent to which the coaches follow the manuals. The aim of this study was to evaluate how cricket coaches understand and coach the batting backlift technique (BBT). This was a mixed-methods research study in which a survey (using both open and closed-ended questions) was electronically distributed among qualified cricket coaches ( $n = 161$ ). These coaches were from eight of the different International Cricket Council playing nations. This study showed that most cricket coaches (83%) coach the SBBT as opposed to the lateral batting backlift technique (LBBT) at various proficiency levels of the game. It was also discovered that towards the higher levels of cricket, most coaches understand the potential value of the LBBT, but have challenges in coaching the LBBT. A recommendation from this study is that cricket national governing bodies should agree on a common method for coaching the BBT in cricket. Since a high level of experience is generally required for coaching a LBBT, further research should be conducted on providing tools and support to assist coaches.

## Introduction

In cricket, an important component of the overall batting technique is the batting backlift technique (BBT). It can be described as a technical component of the batting technique, traditionally described as a movement in a linear plane (Stretch, Bartlett and Davids, 2000; McLean and Reeder, 2000). It has been shown that a majority of successful batsmen in cricket adopt a lateral batting backlift technique (LBBT) (Noorbhai and Noakes, 2016a). It has also been investigated that a LBBT was shown to be more common at the highest levels of cricket (Noorbhai, 2017). However, as one moves through the levels of cricket ability (i.e. junior to adolescent cricket), cricketers adopt more of a straight batting backlift technique (SBBT) (Noorbhai and

Noakes, 2016b). This poses an interesting question of whether the backlift is coached differently between varied proficiency levels.

Since the inception of the game, cricket coaches have been coaching batting techniques to cricketers at various levels. A number of cricket coaching manuals including the Marylebone Cricket Coaching (MCC) manual state that batsmen should lift their bats towards the wicket-keeper or in the direction of the middle stump on the initiation of the batting stroke, with the bat face pointing towards the ground (Beldam and Fry, 1905; Lewis, 1992; The MCC, 1952). This traditional backlift, described as the SBBT, was recommended by one of the first cricket coaches, C.B. Fry, in 1912 (Fry, 1912). Figure 1 summarises some of the most read coaching literature (n = 30) on the recommended direction of the backlift from each coaching book or manual. Only a minority of the most notable cricket coaching manuals (13.3%) advocated that the backlift be directed beyond second slip or in a looped technique (i.e. a LBBT). The remaining coaching manuals advised that the backlift be directed straight towards the stumps or towards first or second slip (i.e. a SBBT 86.7%) (figure 1).

| Author(s), Year                                  | Title                                                      | <b>A:</b> Straight towards the stumps<br>or wicket-keeper<br><b>B:</b> Towards first or second slip<br><b>C:</b> In a looped or lateral<br>technique |  |  |
|--------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| Ranjitsinhji, K.S. (1897)                        | <i>The Jubilee Book of Cricket 5<sup>th</sup> Ed</i>       | A                                                                                                                                                    |  |  |
| Giffen, G. (1898)                                | <i>With Bat and Ball</i>                                   | A                                                                                                                                                    |  |  |
| Grace, W.G. (1899)                               | <i>Cricketing Reminiscences and Personal Recollections</i> | A                                                                                                                                                    |  |  |
| Beldam, G.W. and Fry, C.B. (1905)                | <i>Great Batsmen: Their methods at a glance</i>            | A                                                                                                                                                    |  |  |
| Fry, C.B. (1920)                                 | <i>Cricket: Batmanship</i>                                 | A                                                                                                                                                    |  |  |
| Knight, D.J. (1922)                              | <i>First Steps to Batting</i>                              | A                                                                                                                                                    |  |  |
| Armstrong, W.W. (1924)                           | <i>The Art of Cricket 3<sup>rd</sup> Ed</i>                | A                                                                                                                                                    |  |  |
| Faulkner, G.A. (1926)                            | <i>Cricket: Can it be taught?</i>                          | A                                                                                                                                                    |  |  |
| MacLaren, A.C. (1926)                            | <i>The Perfect Batsman: JB Hobbs in Action</i>             | A                                                                                                                                                    |  |  |
| Jardine, D.R. (1939)                             | <i>Cricket</i>                                             | A                                                                                                                                                    |  |  |
| Wheatley, G.A. Parry, R.H. and Barlee, J. (1948) | <i>Cricket...Do it this way</i>                            | A                                                                                                                                                    |  |  |
| The MCC, (1952)                                  | <i>The MCC Cricket Coaching Book</i>                       | A                                                                                                                                                    |  |  |

|                                                    |                                                                                                         |           |          |          |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------|----------|----------|
| Bradman, D. (1958)                                 | <i>The Art of Cricket</i>                                                                               |           |          | C        |
| Goodwin, C.J. (1967)                               | <i>Coming in to Bat</i>                                                                                 | A         |          |          |
| White, N. and Headley, G. (1974)                   | <i>George 'Atlas' Headley</i>                                                                           | A         |          |          |
| Dellor, R., (1990)                                 | <i>How to Coach Cricket</i>                                                                             | A         |          |          |
| Ferguson, D., (1992)                               | <i>Cricket: Technique, Tactics, Training</i>                                                            | A         |          |          |
| Lewis, T., (1992)                                  | <i>MCC Masterclass: The new MCC Coaching</i>                                                            |           | B        |          |
| Woolmer, B., (1993)                                | <i>Skillful Cricket</i>                                                                                 |           | B        |          |
| Boycott, G. and Gower, D. (1994)                   | <i>Batting Vivian Richards</i>                                                                          |           | B        |          |
| Tyson, F. (1994)                                   | <i>The Cricket Coaching Manual</i>                                                                      | A         |          |          |
| Simpson, B. (1996)                                 | <i>The Reasons Why: A decade of coaching, a lifetime of cricket.</i>                                    | A         |          |          |
| Palmer, D. (1999)                                  | <i>Cricket Coachmaster Batting Mechanics</i>                                                            | A         |          |          |
| Australian Cricket Board, (2000)                   | <i>Coaching Youth Cricket</i>                                                                           | A         |          |          |
| Chappell, G. (2004)                                | <i>Cricket: The making of champions</i>                                                                 |           | B        |          |
| Shillinglaw, T. (2008)                             | <i>Don Bradman's "Continuous 'Rotary' Batting Process"</i>                                              |           |          | C        |
| Shillinglaw, T., (2009)                            | <i>Bradman Revisited 2<sup>nd</sup> Edition 'The Simplicity of Nature'</i>                              |           |          | C        |
| Woolmer, B., Noakes, T.D., and Moffett, H., (2009) | <i>Bob Woolmer's Art and Science of Cricket</i>                                                         |           | B        |          |
| Borooah, V.K. and Mangan, J.E. (2010)              | <i>The "Bradman Class": An exploration of some issues in the evaluation of batsmen for test matches</i> |           |          | C        |
| Woolmer, B. Noakes, T.D. and Moffett, H. (2010)    | <i>Bob Woolmer on Batting</i>                                                                           |           | B        |          |
| <b>TOTAL</b>                                       | <b>30</b>                                                                                               | <b>20</b> | <b>6</b> | <b>4</b> |

**Figure 1:** Summary of relevant coaching literature on the direction or basis of the batting backlift technique in cricket batting

Sir Donald Bradman, who himself was the prime example of this 'looped' BBT (referred to as the LBBT in this study), wrote or was the subject of some of these books (Borooah *et al.*, 2010; Bradman, 1958; Shillinglaw, 2008; Shillinglaw, 2009). These books strongly supported the LBBT technique. However, the majority of the manuals supported the SBBT (Table 1). Despite Bradman's success, one could argue that those in favour of coaching the LBBT have been in the minority – at least as represented in published coaching manuals.

In 2012, Penn and Spratford investigated whether coaching recommendations for batting techniques were supported by findings from biomechanical research. Coaching courses on teaching cricket appear to be less utilised by coaches since the courses do not match the coaches' experiences as well as the teaching content from the coaching courses (Piggot, 2012). This area requires additional research; but the views of experienced coaches and players cannot be excluded, as they provide valuable insights into the reality of teaching the game of cricket by combining practical experiences with the theory. This is especially important in the modern game which is evolving rapidly (Penn and Spratford, 2012; Piggot, 2012; Wulf, 2012; Noorbhai and Noakes, 2015). To complement the evolution of the game, previous research describes two innovative coaching tools that can improve and assist with the coaching of the LBBT.

Firstly, a novel coaching cricket bat was investigated through a pilot and intervention study (Noorbhai and Noakes, 2016c). In the pilot group, cricketers using the coaching bat scored approximately 1 more run per ball or an average of approximately three more runs in total when using the coaching bat than the normal bat ( $F = 6.70$ ,  $df = 1$ ,  $p = 0.012$ ). In the intervention study, the experimental group scored double the total number of runs (an additional 16 runs) and an average of approximately 3 more runs per player in the post-match than in the pre-match, which showed a large effect ( $ES = 5.41$ ) ( $t = 3.32$ ,  $df = 5$ ,  $p = 0.021$ ). It was concluded that a coaching cricket bat might be used to coach young cricket players to hit the ball more effectively as well as how to adopt a more LBBT. The second innovative tool was the development of a mobile application, which can be used by players, coaches and scientists to analyse and improve the BBT among cricketers at all levels (Noorbhai and Noakes, 2016d).

### **Research problem and aims**

As a result, although there is both an understanding of the importance of the BBT in previous studies and how innovative tools of the BBT in cricket could be useful for both cricket players and coaches, there has been no study that has examined whether there are differences in the coaching methods of BBT of cricket coaches at any level. This mixed methods study will address how cricket coaches coach the BBT and to gain any additional insights of their understanding of how different BBT may

influence batting ability in cricket. A convergent parallel mixed methods design will be used, and it is a type of design in which qualitative and quantitative data are collected in parallel, analysed separately, and then merged. In this study, quantitative and qualitative data will be used to explore the central phenomenon and test the theory of the coaching methods of the BBT by cricket coaches at various levels and in various countries. The reason for collecting both quantitative and qualitative data is to obtain different, multiple perspectives, or more complete understandings on the coaching methods and approaches of coaching the BBT in cricket.

We hypothesised that coaches at different levels of the game would coach the BBT in the same manner. In particular, most junior cricket coaches would coach the SBBT whereas towards the higher levels of cricket, most coaches would understand the potential value of the LBBT.

### **Research design**

Mixed methods is a research approach popular in the social, behavioural and health sciences, in which researchers collect, analyse, and integrate both quantitative and qualitative data in a single study to address their research questions (Creswell, 2013). This was a mixed methods study in which a survey was utilised that included both closed-ended questions and an open-ended question. A mixed-methods study combining both qualitative and quantitative analysis is found to be more effective and invaluable for gaining insights from coaches (Knudson, 2007). Therefore, this study is structured on a theory, based on an evaluation gathered from cricket coaches on their coaching methods of BBT in cricket (figure 2).

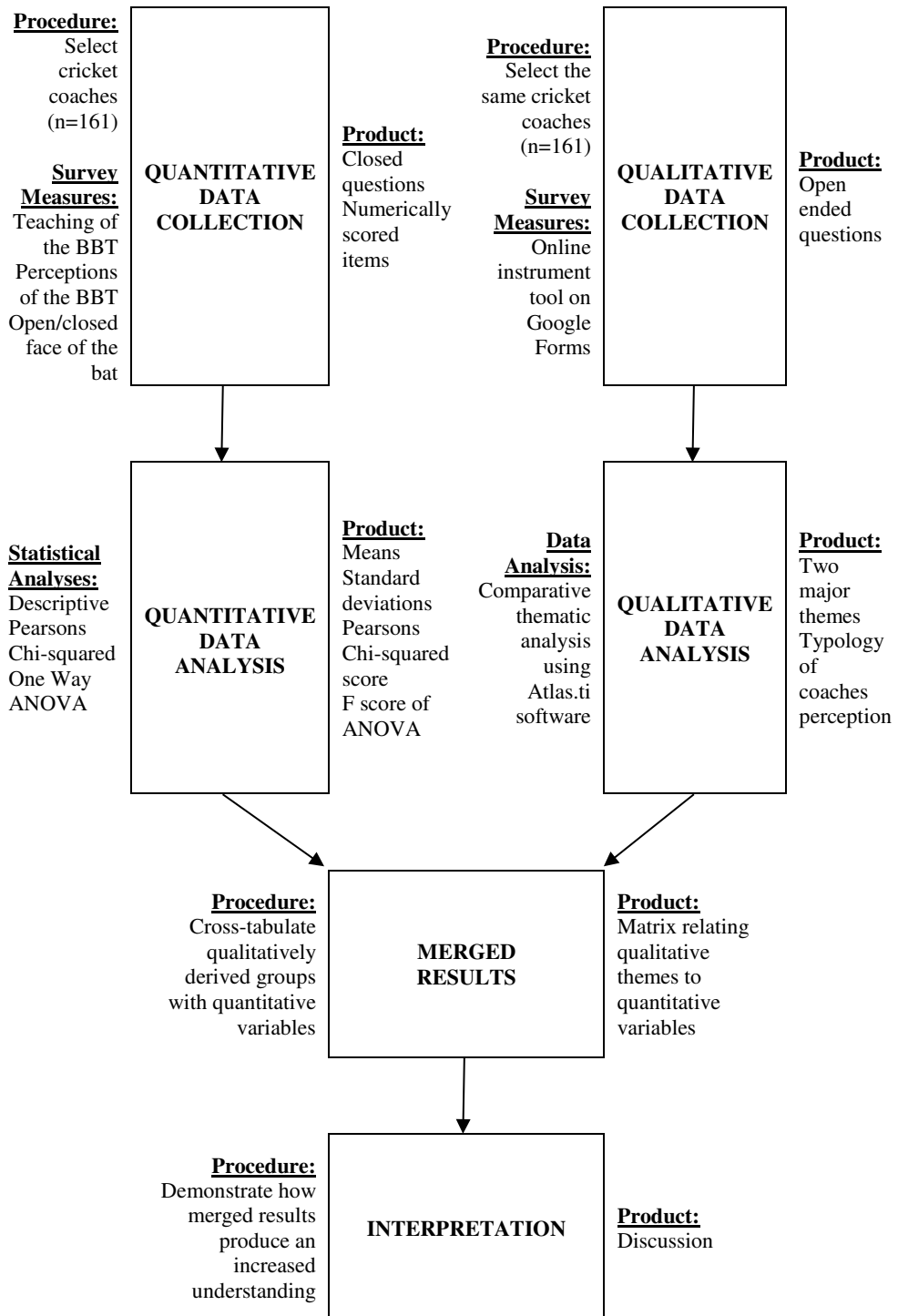
### **Research questions**

Given that most of the coaching manuals and some of the cricket coaching courses advocate the teaching of the traditional SBBT, this theory does not match the practice of the great batsmen who adopt a LBBT (Noorbhai and Noakes, 2016a). The two important questions requiring attention in this study are:

- 1) Are coaches aware of the LBBT?
- 2) What type of BBT are the cricket coaches coaching (if any)?

### **Quantitative data analysis**

Data were captured automatically from GoogleSheets™ and exported onto an online Microsoft Excel sheet upon submission of the coaches' responses. Pearson's Chi-squared tests were conducted to determine any significant trends in the answers obtained from the survey. One-Way ANOVAs were also conducted to determine whether coaches at different levels coach different batting techniques. All statistical analyses were performed using R (R Core Team, 2014) at a significance level of  $\alpha = 0.05$ .



**Figure 2:** Design of mixed methods study to investigate the teachings of the batting backlift technique. (*Adapted from Creswell, 2013*)

## **Qualitative data analysis**

The Atlas.ti Qualitative Data Analysis Software (Scientific Software Development GmbH, Berlin and Germany) was used for analysis for the open-ended data. Once the questionnaire was answered, open-ended answers gathered from the coaches were coded using a thematic coding framework. These codes were formulated from the answers provided by the cricket coaches to the open-ended question. The results are discussed as themes that arose from the quantitative data, which are supported by qualitative data in the form of quotes and insights provided by the cricket coaches.

## **Participants**

All participants (n = 161) were qualified cricket coaches located in eight of the different International Cricket Council (ICC) playing nations (South Africa, Australia, New Zealand, England, India, Pakistan, West Indies and Zimbabwe). These cricket coaches have coached cricket players at various proficiency levels of the game from beginner and junior to elite level.

## **Procedures**

An online survey was utilized, since this was the most appropriate manner by which to reach the participants who were located around the world. The questionnaire was compiled using GoogleForms™. The survey was emailed in the form of a hyperlink (<http://bit.ly/1rgGG3G>) to most of the cricket coaching bodies within each country (this is where Primary school, Secondary School, Club, Junior Provincial and Amateur/State B coaches were reached), and also to individual cricket coaches known to the researcher (mainly franchise and international coaches) who were also asked to disseminate the survey to other cricket coaches in their network. Participants provided consent via the information leaflet prior to the commencement of the survey in which their answers were recorded online. Participants had the option either to remain anonymous or to disclose their names if they wanted to disclose their identity.

To ensure that all the important questions were completed in the survey, questions that had to be answered were denoted with an asterisk (\*). The form could not be submitted without completion of all these required questions. Each survey required between 10-15 minutes for its completion. The questions asked were aimed to elicit cricket coaches' experiences, insights and perceptions of factors promoting and hindering the BBT.

## **The online survey**

The online survey was validated before the questionnaire began, on the basis of what was found in both the previous literature on the backlift, the latest coaching manuals as well as cricket coaching courses. We focused primarily on the coaching manuals since these are the widely preferred sources of coaching information for

coaches, especially up to 2011 (Penn and Spratford, 2012). In addition, most cricket coaching courses offered worldwide provides vague information on the BBT, whereas the cricket coaching manuals and books appear to be much more specific (Gilbert and Trudel, 2001; Cushion *et al.*, 2003; Nelson and Cushion, 2006; Lemyre *et al.*, 2007). The questionnaire began with closed-ended questions. There was an open-ended question towards the end of the survey so that coaches could provide additional comments on the questions asked.

The closed-ended questions included the following:

- Demographics (age, country and level of coaching)
- Do coaches feel that what is advocated in coaching manuals for batting is what happens in actual play?
- Their perception of whether it is important to coach cricket players utilising a generic batting technique.
- Their perception of the value of batsmen using a particular, prescriptive batting technique or adopting their own natural style or ability.
- Their perception of the importance of coaching a cricket player utilising a generic batting technique or as an individual or by applying an approach promoted in a particular cricket book or coaching manual.
- Are they aware of what a 'looped/rotary' technique or lateral backlift is?
- Which backlift do they use to coach cricketers?
- Do they think it is necessary to emphasise the importance of the bat face being open or closed during the backlift?
- What do they advocate in terms of the bat face being either, open, closed or semi-open/closed during the backlift?
- Should the bat face be directed towards the off-side or towards the wicket-keeper/stumps during the backlift?

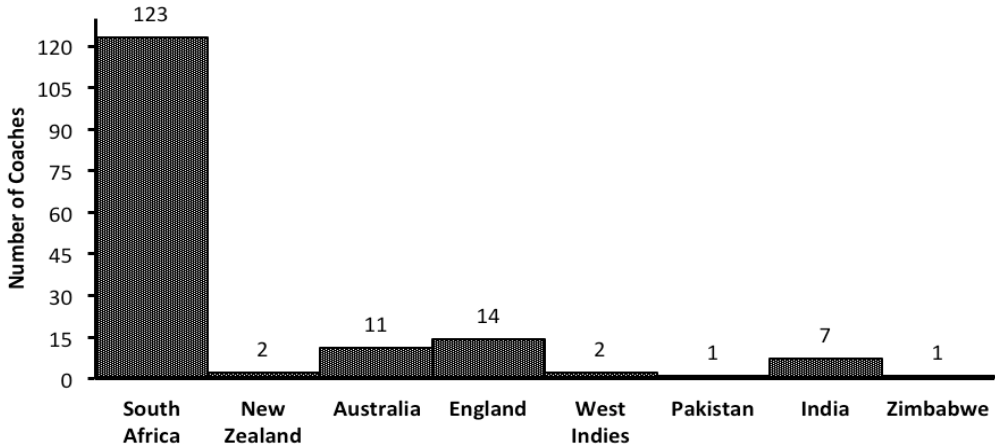
The open-ended question at the end of the questionnaire provided an opportunity for the cricket coaches to explain their individual approach to the coaching of the BBT. Having closed-ended questions might have limited the responses from coaches, but this section was included purposefully in order to analyse the instinctive responses from the cricket coaches based on their current understandings and coaching methods of the BBT.

### **Ethical considerations**

Information leaflets were distributed to participants and informed consent forms were obtained from cricket coaches prior to their participation in the survey. The consent form was presented online as the first page of the questionnaire. Ethical approval for the study was granted by the Human Research Ethics Committee of the University of Cape Town (HREC: 586/2014). This research study conforms to the World Medical Association Declaration of Helsinki on Ethical Principles for Research Involving Human Subjects.

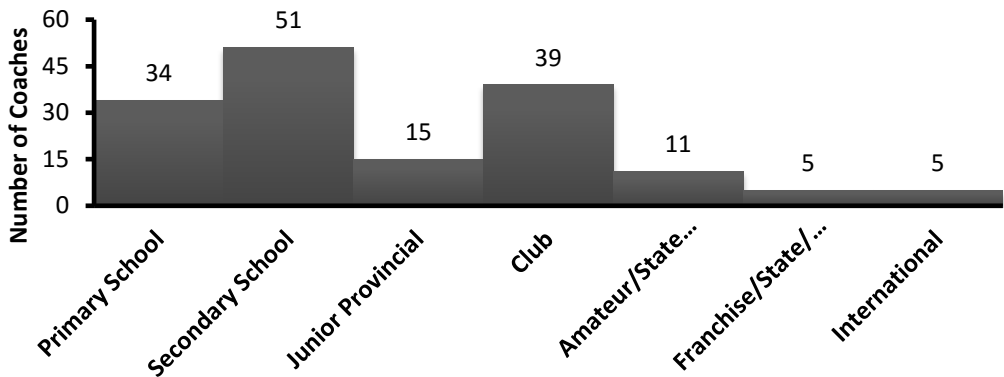
### Results: demographics

The majority of the cricket coaches who participated in this study resided in South Africa (n=123: 77%) and the remaining cricket coaches were from Australia, England, New Zealand, India, Pakistan, West Indies and Zimbabwe respectively (figure 3).



**Figure 3:** Coaches' participation and their country of origin

The response rate was relatively poor from nations outside of South Africa. This posed a challenge given that online communication was the only means for contact with these coaches. Despite the global availability of the questionnaire, it is not immediately clear why the response from South African coaches was so adequate, and with only few responses from coaches in other countries. In addition, a majority of these coaches were found to coach at primary school (21%), secondary school (33%) and club (24%) levels (figure 4). A minority of coaches coached at junior provincial, amateur/state B, franchise/county/state and international level (22%).



**Figure 4:** Number of cricket coaches represented at each level of cricket

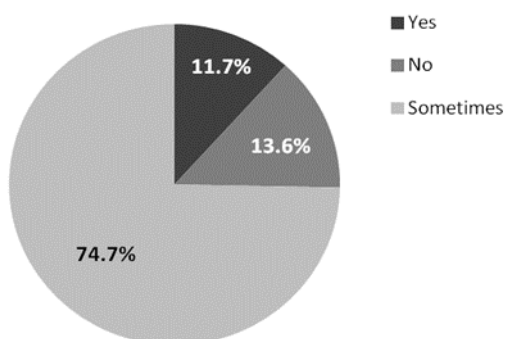
Most of the coaches were in the 26 – 35 and 36 – 45 age categories ( $n = 94$ ; 59%) whereas the remainder were in the 16 – 25 and above the 46-year age categories ( $n = 66$ ; 41%) (figure 5). The sample of coaches in this study had an adequate spread of ages, ranging from 16-25 years to 56+ years.

| Age Category | Number (N) | Percentage (%) |
|--------------|------------|----------------|
| 16 – 25      | 28         | 17.4           |
| 26 – 35      | 58         | 36.6           |
| 36 – 45      | 37         | 22.4           |
| 46 – 55      | 27         | 16.8           |
| 56 – 65      | 9          | 5.6            |
| 66+          | 2          | 1.2            |

**Figure 5:** Age characteristics among cricket coaches who participated in an online evaluation of batting backlift techniques ( $n = 161$ )

### Coaches responses

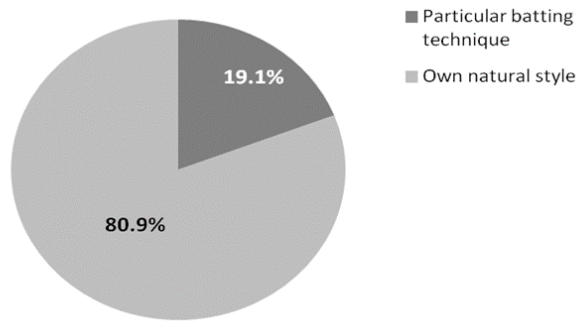
Most of the cricket coaches (74.7%;  $\chi^2 = 9.4$ ,  $p = 0.009$ ) believed that only sometimes are current batting practices accurately reflected in modern cricket coaching manuals. A further 13% of cricket coaches concluded that the current batting practices are not accurately reflected in modern cricket coaching manuals (figure 6).



**Figure 6:** Coaches' response to the following question: Do you think what is advocated in coaching manuals for batting is what happens in actual play?

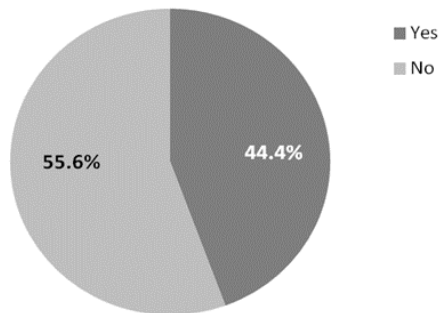
It would therefore seem that the coaches were generally quite skeptical about the extent to which coaching manuals reflect the reality of actual play. With this in mind, we found that a vast majority of cricket coaches indicated that the successful batsmen with whom they work have their own natural style for cricket batting (80.7%;  $\chi^2 = 4.7$ ,  $p = 0.02$ ) (figure 7).

**Figure 7:** Coaches' response to the following question: Do you think successful batsmen use a particular batting technique or their own natural style?



Less than half of cricket coaches however felt that it is not advisable to coach cricket players utilising a generic batting technique (44.4%) (figure 8).

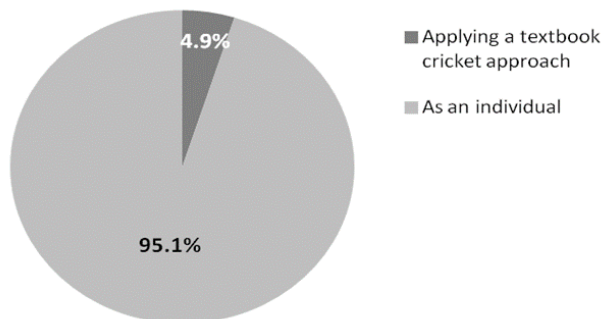
**Figure 8:** Coaches' response to the following question: Do you think it's important to coach each cricket player utilising a generic batting technique?



There was also a considerably high level of consensus that cricket coaches coach a cricket player as an individual and not by applying a textbook cricket approach (95%:  $\chi^2 = 27.7$ ,  $p = 0.0001$ ) (Figure 9).

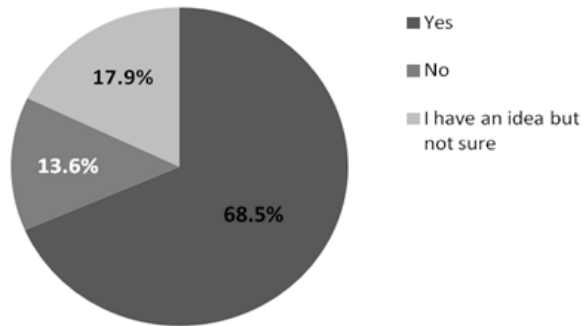
Most of the cricket coaches (65%) indicated in the survey that working with players on an individual basis by applying the basics of batting (some of these were: head still, balance, hitting the ball with the full face of the bat and hitting through the line) is imperative.

**Figure 9:** Coaches' response to the following question: Do you feel it's important to coach each cricket player as an individual or applying a textbook cricket approach?

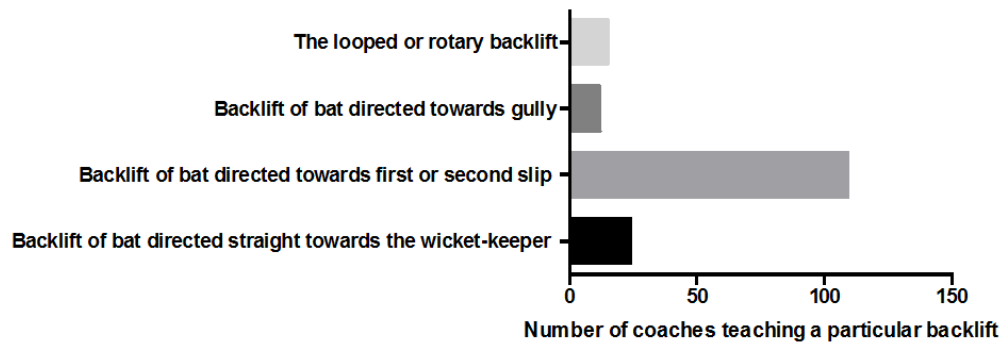


When asked about what a ‘looped’ backlift, LBBT or rotary style of batting, 68.5% of the cricket coaches were aware of this technique (figure 10).

**Figure 10:** Coaches’ response to the following question: Do you know what a ‘looped’ backlift or rotary style of batting is?



Most of the cricket coaches in this study taught the backlift of the bat to be directed towards first or second slip (68%:  $\chi^2 = 10.6$ ,  $p = 0.01$ ) (figure 11).



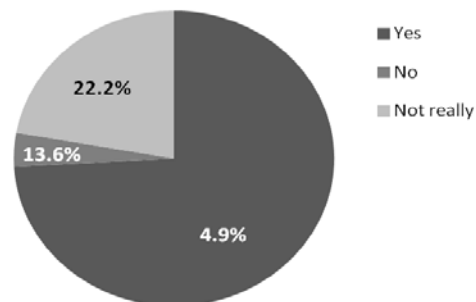
**Figure 11:** Batting backlift technique used by coaches to coach cricketers

|                   | N   | Wicket-keeper (%) | First or second slip (%) | Gully (%) | Looped / Rotary (%) |
|-------------------|-----|-------------------|--------------------------|-----------|---------------------|
| Primary           | 34  | 18                | 65                       | 15        | 3                   |
| Secondary         | 51  | 20                | 69                       | 6         | 6                   |
| Junior Provincial | 14  | 0                 | 71                       | 0         | 29                  |
| Club              | 39  | 10                | 69                       | 8         | 13                  |
| Amateur           | 11  | 18                | 64                       | 18        | 0                   |
| Franchise         | 6   | 17                | 83                       | 0         | 0                   |
| International     | 6   | 0                 | 67                       | 0         | 33                  |
| Total             | 161 | 14                | 68                       | 8         | 9                   |

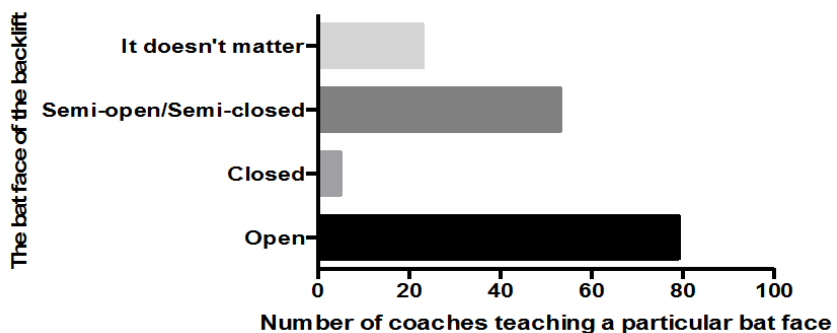
**Figure 12:** Percentage of coaches at various levels coaching batting backlift techniques in different directions when coaching cricketers.

Across all levels, the majority of coaches (64% - 83%) coach the bat to be directed towards the slips (Table 5.3). Significantly, more coaches on average across all levels coach the SBBT as opposed to the LBBT ( $F = 7.28$ ,  $df = 3$ ,  $p = 0.001$ ). A further 10% -20% of coaches at almost every level also coach the bat to be directed towards the wicket-keeper (figure 12). Based on previous research conducted by Noorbhai and Noakes (2016a), a bat directed towards the wicket-keeper or first/second slip is classified as a SBBT. Verily, in practice, most coaches in this study (67% - 100% across all levels) are coaching the SBBT (82% overall), with the vast majority of coaches at the Secondary school level (83%) and all coaches at Franchise level coaching the SBBT. Since a vast majority of coaching manuals advocate for the SBBT, coaches in this study coach what is advocated in the coaching manuals. However, the average number of coaches coaching the SBBT across all levels is not significantly different from those coaching the other techniques ( $F = 3.143$ ,  $df = 2$ ,  $p = 0.06$ ). Aside from the direction of the backlift and other components of the batting technique, this research study also reflected that majority of the cricket coaches (74.5%;  $\chi^2 = 16.3$ ,  $p = 0.0003$ ) (figure 13) emphasise the importance of the bat face being open. The face of the bat however, is not emphasised in coaching manuals.

**Figure 13:** Coaches' response to the following question: With the backlift, do you think it is necessary to emphasise the importance of the bat face being open or closed?



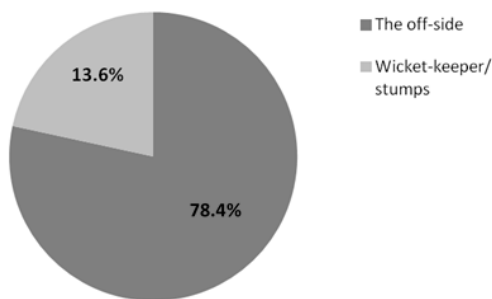
Nearly half of all cricket coaches coach the open face of the bat (49%) and a third of cricket coaches coach the semi-open or semi-closed face of the bat (figure 14). Only a number of cricket coaches in this study coach a closed face of the bat.



**Figure 14:** Number of coaches advocating if the bat face should be open, closed or semi-open/closed

A majority of the cricket coaches (78.4%) felt that the face of the bat should be directed towards the off-side, not towards the wicket-keeper or the stumps (figure 15).

**Figure 15:** Coaches' response to the following question: Should the bat face of the bat be directed towards the off-side or towards the wicket-keeper/stumps?



## Discussion

The process of learning to coach has been subject to scrutiny over the past 20 years. Despite the increase in the number of coach education programmes implemented worldwide, Nelson *et al.*, (2006) argues that 'our understanding of coach learning and the acquisition of professional knowledge lacks a clear conceptual base'. Limited research does however suggest that coach learning is influenced by a mix of formal, non-formal directed and self-directed learning experiences developed by chance (Nelson *et al.*, 2006). It is argued that research has been more influenced by personal and methodological interests of scholars rather than an attempt to develop a conceptual framework as in the case of batting in cricket (Lawson, 1991). We will however draw on Coombs and Ahmeds's framework of formal, non-formal and informal learning as a basis for this discussion of results (Coombs and Ahmed, 1974; reviewed in Nelson *et al.*, 2006).

The main finding of this study showed that coaches tend to coach what is advocated in coaching manuals. This is supported by Coombs and Ahmeds (1974) framework of self-directed informal learning in which learning occurs beyond dedicated formal learning institutions such as by engagement of coaching manuals (Nelson *et al.*, 2006). This finding was drawn from the questions enquiring about the link between coaching manuals and practice as well as what type of backlift the coaches are coaching players. Coaches resist the idea that their coaching approaches are restricted by manuals, but also indicate that they coach the individual and do not adhere to generic coaching guidelines. A secondary finding of this study showed that coaches tend to coach more of the SBBT as opposed to the LBBT at all proficiency levels. This is expected since coaches tend to coach from coaching manuals, which has focused on the coaching of the SBBT (refer to Table 1). This was drawn from the questions enquiring about their knowledge of the looped backlift, their approach to the direction of the backlift and the emphasis of the face of the bat, being open, semi-open/closed or closed. These two main findings are further discussed below.

### **The link between coaching manuals, practice and the individual player**

Coaches felt that the game has changed over the years, and formalising a specific technique may not be ideal for most purposes. This is supported by Renshaw *et al.*, (2010) who states that a key feature in enhancing performance is variability. In addition, coaches felt that using the coaching manuals for players at a younger level provides a foundation for the acquisition of skills. When players are older, beyond their teenage years, coaching manuals would not provide some of the answers that may be paramount for the player. Research demonstrates that through informal learning (such as coaching manuals, video footage, science videos), coaches are able to develop strategies to overcome the above practical coaching dilemmas (Gilbert and Trudel, 2001). Therefore, this method of coach learning should not be underestimated.

One must remember that the game has changed drastically over the years and coaching textbooks have not. A coach should endeavour to realise this and continue to develop his coaching structures. – South African Secondary School Cricket Coach (40.4% of coaches shared the similar sentiment).

According to Parkin (2003), each coaching structure may change over time but it must also have a strong foundation. Some coaches advocate for a Humanistic approach that involves focusing on all aspects of the player, while concentrating on a few key components of technique (Lyle, 2002). From a different perspective, it is perhaps unsurprising that one of the few consistent and global findings in the coaching literature is that coaching knowledge and practices, among both elite and non-elite coaches, are conceptualised from both formal and informal sources (Gilbert and Trudel, 2001; Cushion *et al.*, 2003; Nelson and Cushion, 2006; Lemyre *et al.*, 2007).

Cricket is evolving rapidly and this poses another challenge for coaches. The challenge is that coaches feel that some of the coaching manuals are outdated and not relevant to their current practice of cricket coaching for individual players. It can therefore be argued that these coaches rely on their experiences, which is the most important facet in the development of coaches as noted by Nelson *et al.*, (2006).

I feel that you stick to the textbook as much as possible but adapt to suit each individual. I feel some of the things in the textbook is outdated and therefore I change and adapt certain things so that it works for the individual – South African Primary School Cricket Coach (31.2% of coaches shared the similar sentiment).

It was found that cricket coaches in this study consider the importance of individuality with a player, which supports adopting the athlete-centred approach (Weissensteiner *et al.*, 2008). However, a majority of coaches still apply the generic approach to young players and only later adopt a tailored approach to players as they age and develop.

Cricket is a game of technique and each individual will have his own style which will suite his style of play. As coaches we can only coach the basics and enhance each individual in the various styles until he finds one that would suit his game – South African Secondary School Cricket Coach (18.6% of coaches shared the similar sentiment).

Similarly, most of the South African cricket coaches in this study indicated that players should apply a textbook cricket approach.

I feel that all batsmen should be taught the textbook style to the extent that they can then decide as to whether or not they feel more comfortable with their own natural techniques – South African Secondary School Cricket Coach.

Some cricket coaches even felt that both the individual and textbook cricket approach is advisable which aligns with the global findings noted above.

I feel it's important to both coach each player as an individual and applying a textbook cricket approach. The combination of both is the recipe to success – Zimbabwe Franchise Cricket Coach (6.2% of coaches shared the similar sentiment).

In contrast, few of the coaches in other countries such as India, Pakistan and the West Indies feel that coaching methods and approaches towards batting techniques are also slightly different.

Correct batting techniques are created by batsmen on the field and not by the textbooks or coaching laboratories. However, techniques are dynamic in nature, keep analysing it and apply it as per need and acceptability of individual batsmen – India Franchise Cricket Coach (3.8% of coaches shared the similar sentiment).

In the game of cricket there is always a chance for players to improve and coaches need to find a way to distinguish between the over and under-utilisation of coaching manuals. Textbooks have limited use once the player advances to higher levels, however the utility of coaching manuals for the basic fundamentals of batting is imperative. The emphasis on enhancing natural ability is also important, especially with regards to the BBT (Weissensteiner *et al.*, 2008). Of note are models of batting that have been proposed such as by Bennie and O' Connor (2011). Weissensteiner *et al.*, (2009) developed a model of expertise in cricket batting in which a favourable socio-developmental environment provides the essential foundation for the development of positive psychological attributes, technical skill mastery and superior visual-perceptual skill. Furthermore, Renshaw *et al.*, (2010) propose a constraint-led approach in conjunction with the athlete-centred approach (individuality noted above) to coaching cricket, which is a suitable theoretical method that coaches and scientists can utilise to underpin learning design. By adopting this approach, coaches understand that performance problems can be solved in a number of ways (such as coaching a batting technique) and therefore there is a rejection of the concept of one optimal movement solution.

## Coaching the batting backlift technique

From a coaching and practical perspective, when coaches were asked about what a ‘looped’ backlift or rotary style of batting is, 68.3% of the cricket coaches were aware of this technique. However, most of the cricket coaches in this study taught the backlift of the bat to be directed towards the keeper or towards first or second slip (83%). The key question here is that if most of the coaches are aware of the ‘looped’ or ‘rotary’ technique and if the majority of the coaches coach the backlift towards the wicket-keeper or the slips, why are they not coaching the ‘looped’ backlift? One explanation is that some coaches perceive the LBBT to be more difficult for batsmen to master than the SBBT. This also makes the technique more challenging to coach.

A coach requires a lot of experience if coaching a backlift beyond first or second slip. It is very clear that a majority of young players attempting the rotary style of batting cannot execute it, hitting across the line - Australian International Coach (4.1% of coaches shared the similar sentiment).

I believe there are some benefits to be gained from using the rotary technique particularly with regard to bat speed and hitting power. In my experience however this technique is more difficult to master for some players and is perhaps most suited to batsmen who have a greater natural ability to time the ball – South African Club Cricket Coach (9.4% of coaches shared the similar sentiment).

Although this finding does not provide an indication as to whether the coaches are comfortable to coach the LBBT or not, the path to coaching expertise is indeed not well understood (Grundel *et al.*, 2013; Nelson *et al.*, 2006). Furthermore, the change of thinking (coaching the LBBT) is perhaps one of the greatest challenges for coaches as well (Renshaw *et al.*, 2010). Aside from the relevance of coaching expertise, some of the cricket coaches believed that the coaching of the backlift is not as important as other elements of the batting technique.

The backlift by itself isn't as important as where the face is pointing at the point of contact. The "traditional" approach is likely to be best for cricketers just learning, as it presents the face forward throughout the stroke, but as long as the batsman is presenting the face of the bat to the ball at contact, how it actually gets there is secondary - England Club Cricket Coach (6.8% of coaches shared the similar sentiment).

This generalisation may certainly be true, but not all strokes in cricket are played with the full face of the bat unless a batsman is driving a ball straight down the ground (this area in cricket is referred to as the ‘V’) (Woolmer *et al.*, 2010). In essence, it has been shown that batsmen can still play most cricket strokes with the full face of the bat if a LBBT is employed (Noorbhai, 2017).

A few of the coaches in this study felt that it would be adequate to emphasise the face of the bat being open, semi-open/closed or closed.

The current coaching recommendation of open bat face coming down from first slip is adequate for most purposes – Australian International Coach (5% of coaches shared the similar sentiment).

In contrast, few of the coaches in this study felt that it would not be important to emphasise the face of the bat being open, semi-open/closed or closed.

The backlift is in any case a means of generating power, the face of the bat can be unimportant as long as the point of contact is correct – South African Secondary School Cricket Coach (6.2% of coaches shared the similar sentiment).

This perception probably supports the traditional approach over the years as to why coaching manuals such as the MCC did not emphasise on the importance of the face of the bat because it is in fact, relatively unimportant which way the bat is angled (The MCC, 1954). In addition, this approach might be adequate for players not struggling, but can pose a challenge for such players who are encountering challenges in their batting earlier on in their careers. Bearing in mind the individuality of coaching players, some of the coaches adopted an approach that works best for the player (athlete-centred approach) (Weissensteiner *et al.*, 2009).

Each player has a unique approach to batting. If the kid is scoring runs and not being dismissed in a "similar" way each game I will leave him/her as is. Some batters have a natural loop without me telling them and some use the conventional backlift, I would opt on working with the batters individually instead of dictating how they should go about the backlift – South African Junior Provincial Cricket Coach (24.8% of coaches shared the similar sentiment).

This notion can be further supported by Beek (2000) in that the natural way of learning movement skills such as batting takes place at a sub-conscious level and cannot be enforced on to players via explicit instructions. However, Lyle (2002) argues that there is little empirical evidence to suggest that one coaching style is more effective than another and that a coaching style still has to have balance and substance. In addition, in cricket, coaches tend to judge the ability of players based on how they perform in the nets or practice without realising the variance of player types (match player, net player or dual players). Coaching that takes place in the nets should only provide an indication to the coach of the players' potential and not his/her skill level, as this can only be accurately determined during matches (Renshaw *et al.*, 2010; Portus and Farrow, 2011).

A flat backlift with a straight bat will look great in the nets and avoid failure making the coach feel great. This however, makes the player succeed and ingrain a process which does not meet the same requirements as the game – England International Cricket Coach (3.1% of coaches shared the similar sentiment).

Based on the above insights, one can notice that coaches have been coaching more of the SBBT (83%) as opposed to the LBBT (17%) at various levels. More so, it is noticeable from this study that most coaches towards the higher levels of cricket understand the potential value of the LBBT (bat directed beyond second slip or as a 'looped' technique) but potentially experience difficulty in coaching it.

### **Recommendations**

The recommendation from this study is that coaches should be educated on how to coach the LBBT since variability is a key feature in enhancing performance (Renshaw *et al.*, 2010). Cushion *et al.*, (2003) suggests that collective understandings begin to develop through experiences and as such, the culture of coaching the LBBT can start to take shape. Since a large proportion of literature on coach learning has tended to focus on expert coaching practitioners, there has been little appreciation of the teaching and learning preferences of coaches across developmental spectrums. The development of tools to support and assist coaches is integral so that they can feel more confident and competent to coach the LBBT, especially among the junior levels.

### **Strengths and limitations**

The first strength of the study was the adequate sample size ( $n = 161$ ). Another strength of the study was the ability to have a diverse spread of cricket coaches based on their age and the proficiency level at which they are coaching. The coaches' varied ages indicate that they have varying levels of experience, which possibly may affect their coaching methods and approaches to the BBT. A limitation of this study was that majority of the cricket coaches in the study sample were South African and there was a low representation of cricket coaches from the other countries, which makes the findings more representative from a South African perspective.

A further limitation of the study was that most of the questions asked on the survey were closed-ended and not open-ended questions. Having closed-ended questions might have limited the responses from coaches. However, the inclusion of one open-ended question allowed for the capturing of additional insights, comments and suggestions by the cricket coaches. The findings from this study have contributed towards foundational insights from coaches on the coaching methods of the backlift in cricket. As such, an in-depth qualitative evaluation is therefore required to explore specific insights and perceptions of the overall batting technique in cricket among cricket coaches at various levels.

### **Further qualitative research**

Based on the above findings, further studies should consider investigating the coaching methods of the BBT by coaches through individual interviews and focus group discussions consisting of open-ended questions. Examples of questions to be asked are:

- If a young player was struggling with their backlift, what would your coaching approach be and why?
- Seeing that coaching should emphasise more on individual ability, on what basis would you coach a LBBT?

## Conclusion

This study has shown that a majority of cricket coaches coach what is advocated in cricket coaching manuals. This study also showed that cricket coaches mostly coach the SBBT as opposed to the LBBT at various levels of cricket ability. It has been highlighted that some experience is generally required if coaching a player to lift the bat beyond second slip or as a LBBT. Further qualitative research is required in order to evaluate the coaching methods of cricket coaches for other elements of the batting technique (grip, stance, downswing, impact with the ball and follow through). The findings from this research study suggests that all various coaching cricket bodies worldwide should meet in order to link the science and coaching of the BBT. This would allow for coaches to be clear and consistent on the coaching of batting techniques (specifically on the BBT) among cricketers at all levels.

## References

- Beek, P.J. (2000) Toward a theory of implicit learning in the perceptual motor domain. *International Journal of Sport Psychology*, 31, 547-554.
- Bennie, A. and O'Connor, D. (2011) An effective coaching model: The perceptions and strategies of professional team sport coaches and players in Australia. *International Journal of Sport and Health Sciences*, 9, 98-104.
- Beldam, G.W. and Fry, C.B. (1905) *Great batsmen: their methods at a glance*. The Macmillan Company, New York
- Borooah, V.K. Mangan, J.E. (2010) *The Bradman Class: An exploration of some issues in the evaluation of batsmen for test matches, 1877-2006*. Berkeley Electronic Press, Berkeley.
- Bradman, D. (1958) *The art of cricket*. Robson Books Ltd, Hodder and Stoughton, London.
- Chappell, G. (2004) *Cricket: the making of champions*. Lothian Books, Victoria.
- Coombs, P.H. and Ahmed, M. (1974) *Attacking rural poverty: how nonformal education can help*. Johns Hopkins University Press, Baltimore.
- Cushion, C., Armour, K. and Jones, R. (2003) Coach education and continuing professional development: experience and learning to coach. *Quest*, 55, 215-230.
- Fry, C.B. (1912) *Cricket: batsmanship*. Eveleigh Nash, London.
- Gilbert, W. and Trudel, P. (2001) Learning to coach through experience: reflection in model youth sport coaches. *Journal of Teaching and Physical Education*, 21, 16-34.
- Knudson, D. (2007) Qualitative biomechanical principles for application in coaching. *Sports Mechanics*, 6, 1, 109-118.
- Lawson, H. (1991) Future research on physical education teacher education professors. *Journal of Teaching in Physical Education*, 10, 229-248.

- Lemyre, F., Trudel, P. and Durand-Bush, N. (2007) How youth-sport coaches learn to coach. *The Sport Psychologist*, 21, 191-209.
- Lewis, T. (1992) MCC Masterclass: The new MCC Coaching. Orison Publishing, London.
- Lyle, J. (2002) *Sports coaching: a framework for coaches' behaviour*. Routledge, London.
- McLean, S.P. and Reeder, M.S. (2000) Upper extremity kinematics of dominant and non-dominant side batting. *Journal of Human Movement Studies*, 38, 4, 201-212.
- Nelson, L., Cushion, C. and Potrac, P. (2006) Formal, non-formal and informal coach learning. *International Journal of Sports Science and Coaching*, 1, 3, 247-259.
- Noorbhai, M.H. and Noakes, T.D. (2015) Advancements in cricket in the 21<sup>st</sup> century: science, performance and technology. *African Journal for Physical, Health Education, Recreation and Dance*, 21, 4:2, 1321-1331.
- Noorbhai, M.H. and Noakes, T.D. (2016a) A descriptive analysis of batting backlift techniques in cricket: Does the practice of elite cricketers follow the theory? *Journal of Sports Sciences*, 34, 20, 1930-1940.
- Noorbhai, H. and Noakes, T.D. (2016b) An analysis of batting backlift techniques among coached and uncoached cricket batsmen. *South African Journal for Research in Sport, Physical Education and Recreation*, 38, 3, 143-161.
- Noorbhai M.H., Woolmer, R.C. and Noakes T.D. (2016c) A novel coaching cricket bat: can it be used to enhance the backlift and performance of junior cricket batsmen? *BMJ Open: Sport and Exercise Medicine*, 2, 1, e000141.
- Noorbhai M.H., Chhaya, M.M.A. and Noakes T.D. (2016d) The use of a mobile application to predict the batting backlift technique among cricket batsmen. *Cogent Medicine: Sport and Exercise Science*, 3, 1, 1214338.
- Noorbhai M.H. (2017) *The batting backlift technique in cricket*. PhD Thesis. University of Cape Town, South Africa.
- Palmer, G. (1999) *Cricket coachmaster batting mechanics*. The Baulch Group, UK.
- Penn, M.J. and Spratford, W. (2012) Are coaching recommendations for cricket batting technique supported by biomechanical research? *Sports Biomechanics*, 11, 3, 311-323.
- Piggott, D. (2012) Coaches' experiences of formal coach education: a critical sociological investigation. *Sport, Education and Society*, 17, 4, 535-554.
- Portus, M.R. and Farrow, D. (2011) Enhancing cricket batting skill: implications for biomechanics, skill acquisition research and practice. *Sports Biomechanics*, 10, 4, 294-305.
- Renshaw, I., Chow, Y., Davids, D. and Hammond, J. (2010) A constraints-led understanding of skill acquisition and game play: a basis for integration of motor learning theory and physical education praxis? *Physical Education and Sport Pedagogy*, 15, 2, 117-137.
- Shillinglaw, T. (2008) *Don Bradman's continuous rotary batting process*. Sage, London.
- R Core Team. (2014) *R: A language and environment for statistical computing*. R Foundation for Statistical Computing, Vienna, Austria [online]. Available at: <http://www.R-project.org> (Accessed 25<sup>th</sup> June 2018).
- Stretch, R., Bartlett, R. and Davids, K. (2000) A review of batting in men's cricket. *Journal of Sports Sciences*, 18, 12, 931-949.
- The MCC (1952) *The MCC Cricket Coaching Book*. The Naldrett Press, London.
- The MCC. (1954) *The MCC Cricket Coaching Book*. William Heinemann Ltd. London.

Tyson, F. (1976) *Complete cricket coaching*. Pelham Books Ltd. Australia.

Weissensteiner, J., Abernethy, B., Farrow, D., Müller, S. (2008) The development of anticipation: a cross-sectional examination of the practice experiences contributing to skill in cricket batting. *Journal of Sport and Exercise Psychology*, 30, 663-684.

Weissensteiner, J., Abernethy, B., and Farrow, D. (2009). Towards the development of a conceptual model of expertise in cricket batting: a grounded theory approach. *Journal of Applied Sport Psychology*, 21, 3, 276-292.

Woolmer, B. (1993) *Skilful cricket*. A and C Black, London.

Woolmer, B., Noakes, T.D. and Moffett, H. (2009) *Bob Woolmer's art and science of cricket*. Struik Publishers, Cape Town, South Africa.

Woolmer, B., Noakes, T.D., Moffett, H. (2010). *Bob Woolmer on Batting*. Struik Publishers, Cape Town, South Africa.

## JQRSS Author Profiles

**Habib Noorbhai**<sup>1</sup> is a researcher in Sports Science and an academic consultant at the Department of Human Movement Science at the University of Fort Hare (SA). His research interests range from cricket science, sports performance and sports nutrition. He has also published scientific papers and presented at conferences in these areas. He completed a BA in Sport Psychology (UJ), Honours in Biokinetics (UKZN), MPhil in Biokinetics (UCT) and a PhD in Exercise Science at UCT. In 2015, he was nominated among Mail and Guardians top 200 young South Africans and in 2017, won the coveted title of Mr South Africa.

**Tim Noakes**<sup>2</sup> studied at the University of Cape Town (UCT) and obtained an MBChB degree in 1974, an MD in 1981 and a DSc (Med) in Exercise Science in 2002. Professor Noakes has published more than 750 scientific books and articles and has been cited more than 16,000 times in scientific literature, has an H-index of 71 and has been rated an A1 scientist by the National Research Foundation of South Africa for a second 5- year term. He has won numerous awards over the years and made himself available on many editorial boards. Professor Noakes is also the founder of The Noakes Foundation.

## Reviewer Comments

This paper has attempted to quantify what is essentially a highly personalised aspect of batting technique. The Creswell model has provided rich variety in data collection strategies allowing triangulation of findings, strengthening the research. A discussion point to further the investigation could be if some field observations of the back-lift actions were made to complement the opinions of coaches from either interviews or books. A key outcome was that elite performers may not heed coaches' advice and develop their own technique, however, some earlier literature about the formation of batting technique would find that pioneers of the game such as the Grace brothers initiated the use of the front foot, which may have brought about the necessity for a back-lift to improve batting. Additionally, some of the sports best batsmen tend to lower their eye line to pick up the ball; crouching in effect, which is also not promoted in the coaching manual, so may strengthen the argument of this paper in that performance is not necessarily coach driven but developed by the elite batsman.